Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *French Hospital Clinical NLP — Metropolitan France (DrBenchmark)*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output Content was flagged HIGH priority due to the deployment's Silver Standard framing and the explicit gap between academic NLP annotators and practicing PMSI coders/hospital clinicians. Findings strongly confirm this concern. Annotation provenance varies dramatically across the 20 datasets and is generally undocumented for clinical-coding authority. DiaMed is the only dataset with explicit IAA statistics and medical expert involvement (one expert) [Q45, Q46]; CAS POS tags are silver-standard automatic via Tagex [Q42, CAS-D15]; ESSAI POS tags are automatic via TreeTagger [Q43]; DEFT-2021's MeSH multi-label and NER annotations are described as 'manually annotated' but annotator credentials are not documented [Q25, DEFT2021-D16]; CLISTER STS scores are averaged across multiple annotators of unspecified expertise [Q41, CLISTER-D12]. Web-source confirmation: no PMSI coder or hospital coding specialist involvement is documented in any DrBenchmark constituent dataset. Multiple dataset-level findings show this matters: CLISTER-D12 ('lost to follow-up' vs. 'death at J13' rated as 1.0 similarity — clinically critical distinction underweighted by non-specialist annotators); CLISTER-D13 (clinical abbreviation 'UB' likely Buruli ulcer requires domain expertise); E3C-D11, E3C-D12 (identifiable pathogens like Nocardia asteroides and treatments left unlabeled — annotation incompleteness); CAS-D17 (annotation rationale opaque for ambiguous speculation labels); ESSAI-D14 (consecutive all-zero NER sequences despite clinically relevant content). DiaMed's annotation is further compromised for the metropolitan deployment by sub-Saharan/North African case origin (DIAMED-D7, DIAMED-D8, DIAMED-D9) — the labels reflect Pan African Medical Journal cases, not metropolitan French hospital coding. Author affiliations are exclusively academic (Nantes Université, Avignon Université, INSERM, CNRS, Université de Lille) [Q7, Q8] with no PMSI coder or hospital coding specialist representation. The deployment's explicit annotator-vs-user gap concern is fully confirmed.

ID	Evidence
Q7	“Yanis Labrak, Adrien Bazoge, Oumaima El Khettari, Mickael Rouvier, Pacôme Constant dit Beaufils, Natalia Grabar, Béatrice Daille, Solen Quiniou, Emmanuel Morin, Pierre-Antoine Gourraud and Richard Dufour” (p.1)
Q8	“LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille” (p.1)
Q25	“This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER.” (p.3)
Q41	“CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided.” (p.4)
Q42	“CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively.” (p.4)
Q43	“ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus.” (p.4)

Q45	“DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session.” (p.4)
Q46	“Table 4: Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1.” (p.4)
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique … keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
E3C-D11	"La culture a isolé le germe Nocardia asteroides." all tags [0,0,0,0,0,0,0,0,0] — "Nocardia asteroides" (pathogen) not tagged as CLINENTITY — suggests annotation incompleteness
CLISTER-D12	Le patient a été perdu de vue." vs. "Le patient décède à J13. — Outcome divergence (lost to follow-up vs. death on day 13); clinically significant difference potentially underweighted by non-specialist annotators
E3C-D12	"Le patient a été mis sous traitement par ciclosporine avec une évolution rapide vers une leucémie aigue myéloblastique." all tags [0,...,0] — Clear pathology and treatment entities untagged — annotation completeness concern
CLISTER-D13	Un patient de 42 ans, a été hospitalisé pour un UB de la main droite. — Clinical abbreviation "UB" (likely Buruli ulcer); domain expertise needed for reliable similarity annotation
ESSAI-D14	"avec la combinaison gemcitabine + abraxane, chez des patients avec un cancer du pancréas" (and 9 similar) — Consecutive all-zero NER sequences despite clinically relevant content — sparse annotation density
CAS-D15	annotated for POS tagging with 31 classes using automatic annotations through Tagex 3...yielded 98% precision — Silver-standard label provenance for POS config
DEFT2021-D16	(see D6 excerpt) — Multi-drug cardiac case; annotator PMSI credential unknown — label authority concern
CAS-D17	`"Cholstat ® 0.1 ."` — Drug name/dosage fragment labeled as speculation with no apparent epistemic marker; annotation rationale opaque

Strengths

- DiaMed provides documented IAA with Cohen's Kappa and Gwet's AC1 across two annotation sessions and explicit medical-expert involvement [Q45, Q46]
- DEFT-2021 is described as manually annotated for both MLC and NER tasks [Q25]
- CAS Tagex automatic POS annotations were validated against manual annotations at 98% precision [Q42] — partial quality assurance
- Benchmark authors include clinical-affiliated institutions (CHU Nantes, Service de Neuroradiologie, l'institut du thorax) [Q8] indicating clinical proximity even if not PMSI-coder authority

ID	Evidence
Q8	“LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, UMR 6004, Service de Neuroradiologie diagnostique et interventionnelle, l'institut du thorax, UMR 8163 – STL CNRS, Université de Lille” (p.1)
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Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect practicing PMSI coder or hospital clinical coding specialist perspectives. The single documented medical expert is in DiaMed [Q45], whose cases are non-metropolitan (DIAMED-D7, DIAMED-D8). Labels reflect academic NLP annotator judgment, not the operational hospital coding population whose authority the deployer requires.

OC-2: Significant disagreement potential. Examples: CLISTER-D12 (clinically distinct outcomes 'lost to follow-up' vs. 'death' rated as low similarity but clinical significance is asymmetric); CLISTER-D13 ('UB' abbreviation requires domain expertise non-specialist annotators may lack); E3C-D11, E3C-D12 (pathogens and treatments left unlabeled — practicing clinicians would likely tag these). Documented PMSI coding quality concerns [WEB-19, WEB-20] independently confirm that even within practicing coder populations, ICD coding has documented inter-coder variability.

OC-3: Annotator demographics are minimally documented. Author affiliations [Q7, Q8] are academic NLP and clinical research institutions; no Datasheet or Data Statement at annotator level is published for most constituent datasets. CAS uses automatic Tagex annotations [Q42, CAS-D15]. ESSAI uses automatic TreeTagger [Q43]. DEFT-2021 MeSH annotation provenance is described only as 'manually annotated' [Q25, DEFT2021-D16]. CLISTER documents 'several annotators' without expertise breakdown [Q41, CLISTER-D12].

OC-4: Re-annotation by a representative regional pool (PMSI coders + hospital clinicians) is recommended for at least DEFT-2021 (the deployment's most aligned multi-label task) and DiaMed (the only ICD-10-labeled task). The deployment's Silver Standard framing implicitly already plans for clinician adjudication; the benchmark itself does not provide labels validated by this population.

OC-5: Aggregation methods erase minority perspectives. CLISTER averages multi-annotator similarity into a single float [Q41, DEFT2020-D16]; CAS and ESSAI use silver-standard automatic POS tags with no human disagreement signal [CAS-D15]; DiaMed's IAA captures inter-annotator agreement but final labels are not multi-label or confidence-scored (DIAMED-D10). The aggregation discards the very disagreement signal that could inform deployment uncertainty flagging.

OC-6: Convergent validity is harmed: labels do not correlate with PMSI coder or hospital clinician judgments in documented form. External validity is harmed: judgments by academic NLP annotators on edited published cases are not expected to generalize to active hospital coding by certified coders. Documented PMSI coding quality concerns [WEB-19] reinforce that even mapping to practicing-coder-equivalent labels is non-trivial.

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DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
E3C-D11	"La culture a isolé le germe Nocardia asteroides." all tags [0,0,0,0,0,0,0,0,0] — "Nocardia asteroides" (pathogen) not tagged as CLINENTITY — suggests annotation incompleteness
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DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
WEB-19	PMSI-MCO bacterial resistance study: ICD coding quality is unequal and some disease categories cannot be reliably characterized from PMSI coding alone, partly due to initial coding errors (https://pubmed.ncbi.nlm.nih.gov/29673880/)
WEB-20	French PMSI bacterial resistance study (ScienceDirect 2018): documented inconsistencies between infection types, causative agents, and resistance codes attributed to initial coding errors (https://www.sciencedirect.com/science/article/abs/pii/S0399077X17307072)

Information Gaps

- Specific clinical credentials of the 'medical expert' in DiaMed annotation [Q45]
- DEFT-2021 competition annotator demographic breakdown not published in benchmark documentation

Requires Expert Verification

- Empirical inter-annotator agreement between DrBenchmark labels and a sample of PMSI-coder re-annotations on a representative metropolitan French hospital subset

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Remediation

Gap 1: No DrBenchmark dataset documents PMSI coder or hospital coding specialist annotation; only DiaMed has one medical expert; CAS/ESSAI POS tags are silver-standard automatic; CLISTER and E3C show

domain-expertise concerns and annotation incompleteness

Recommendation: Recruit certified PMSI coders to re-annotate a representative subset of DEFT-2021 multi-label cases and DiaMed ICD-10 assignments. Compute Cohen's Kappa / Gwet's AC1 between PMSI-coder labels and benchmark labels to quantify the annotation-authority gap explicitly. Use the disagreement signal as a calibration target for the deployment's uncertainty-flagging behavior.